

Pre-Release Uncertainty and CPI Forecast Errors: Evidence from a Hybrid Kalshi–Adaptive Baseline Panel, 2019–2026

Siddharth Korukonda[†] Jiabin Chen[†] Andrew Luong[†]

[†]Stony Brook University

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Abstract

We study whether a measure of pre-release forecast uncertainty—constructed from Kalshi KX-CPI prediction-market contracts where available, and from a documented error-volatility proxy otherwise—predicts the magnitude of U.S. CPI forecast errors. The panel spans 77 monthly headline CPI observations from August 2019 through February 2026. Ten of those months overlap with live Kalshi contract data pulled through the Kalshi v2 API (155 range contracts across 21 KXCPI events, of which 10 events correspond to already-released CPI months); the remaining 67 months use an adaptive-expectations forecast (EWM of past CPI) as a survey-consensus proxy and that proxy plus an oil/VIX correction as a Kalshi proxy. We are explicit throughout: this is *not* a direct Kalshi-vs.-SPF comparison over the full sample, and we do not present it as one.

Within this hybrid panel, three findings are consistent across model families. First, the Kalshi-implied (and, for non-overlap months, proxy) standard-deviation measure σ_t is positively associated with the absolute CPI surprise $|\epsilon_t|$: OLS with Newey–West standard errors yields $\hat{\beta}_\sigma = 0.718$ ($p = 0.070$, $R^2 = 0.421$, $F_{9,67} = 10.59$), and the relationship amplifies in the upper tail of the error distribution (quantile regression: 1.058 at $\tau = 0.50$, 1.415 at $\tau = 0.75$, 1.515 at $\tau = 0.90$). Second, the unconditional accuracy gap between the Kalshi series and the survey proxy is +0.008 percentage points (Survey MAE = 0.218, Kalshi MAE = 0.210), which is economically small and statistically marginal (Diebold–Mariano $p = 0.074$). Third, contrary to the hypothesis that prediction markets would dominate in turbulent periods, the Kalshi conditional win rate is *lower* in the highest-uncertainty tercile (50.0%) than in the lowest (65.4%).

Block-bootstrap confidence intervals and a 29-specification robustness table indicate that the uncertainty–miss association is directionally stable (positive sign in roughly 25 of 29 specifications) but not uniformly significant; the bootstrapped confidence interval for $\hat{\beta}_\sigma$ includes zero. We therefore read the results as suggestive evidence that a market-based dispersion signal tracks forecast difficulty, particularly in the tail, and as null-to-negative evidence that a prediction-market point forecast dominates a survey-style anchor in high-uncertainty months. Stronger claims require a longer direct-contract sample and real SPF/Bloomberg consensus data.

Keywords: CPI forecasting, prediction markets, Kalshi, forecast uncertainty, adaptive expectations, quantile regression, Diebold–Mariano, time-series cross-validation.

JEL Classification: E37, G14, C53.

1 Introduction

The monthly Consumer Price Index release is one of the most heavily traded macro events on the U.S. calendar. A meaningful forecast miss repositions Treasury yields, recalibrates expectations for the FOMC path, and moves trillions of dollars in rate-sensitive assets. Any systematic way to anticipate when a forecast is likely to fail has direct practical value, independent of whether a better point forecast can be produced.

Historically the dominant pre-release signal has been a survey consensus—Bloomberg, Reuters, or the Philadelphia Fed’s Survey of Professional Forecasters (SPF). Surveys aggregate expert judgment at fixed deadlines. Prediction markets are a structurally different mechanism: participants trade binary contracts whose payoffs depend on the realized outcome, and the resulting price surface evolves continuously in response to news. Kalshi’s KXCPI market exposes this structure particularly cleanly: each monthly CPI event consists of a ladder of binary threshold contracts (“CPI above 0.1%?”, “above 0.2%?”, ...) whose prices form a discretized survival function over possible CPI outcomes. Differencing adjacent contract prices gives an implied probability distribution, from which a point forecast, a standard deviation, an entropy, and tail masses can be recovered without parametric assumptions.

The original question that motivated this project was whether, conditional on elevated pre-release uncertainty, the Kalshi-implied forecast would outperform the survey consensus—a conditional restatement of the standard prediction-market claim that incentive-aligned, continuously-updating mechanisms dominate in information-rich or fast-moving environments. A clean test of that question requires two things this paper does *not* fully have: (i) real-time SPF or Bloomberg consensus for every month, and (ii) direct Kalshi contract data over a long window. KXCPI launched only recently, and over our 2019–2026 window we have live contract data for 21 monthly events, of which only 10 correspond to already-released CPI prints and therefore enter the realized-accuracy analysis.

We take a conservative path. We construct a single, transparent hybrid panel. For the 10 months with realized CPI and live Kalshi data, the implied forecast and uncertainty measures are computed directly from the threshold-to-distribution conversion. For the remaining 67 months, we use a documented adaptive-expectations proxy as a stand-in for a survey anchor, and a small oil/VIX-adjusted version of that proxy as a stand-in for a Kalshi point forecast. The uncertainty measure σ_t is the real implied standard deviation on overlap months and a rolling six-month volatility of prior forecast errors on non-overlap months. Every construction is labelled explicitly throughout.

Within this setup we ask two predictive questions:

1. **Uncertainty as a leading indicator of miss size.** Does σ_t predict $|\epsilon_t|$ in the following release, and does the relationship concentrate in the upper tail of the error distribution?
2. **Conditional relative performance.** In months flagged as high-uncertainty, does the Kalshi series (real-data overlay plus proxy elsewhere) beat the adaptive survey baseline in absolute error?

We formulate three hypotheses:

- H1.** σ_t is positively associated with $|\epsilon_t|$ in the subsequent CPI release.
- H2.** The Kalshi win rate exceeds 0.5 in high-uncertainty months.
- H3.** The unconditional mean error gap $\text{MAE}_{\text{survey}} - \text{MAE}_{\text{kalshi}}$ is positive.

We emphasize throughout that the analysis is predictive, not causal; that the accuracy comparison is, for most months, a comparison of two closely-related constructed series rather than of a real prediction market against a real survey; and that conclusions about the Kalshi point forecast *per se* must be read against the limited direct-contract window. With those caveats, the paper’s contribution is threefold: a transparent, reproducible pipeline for converting KXCPI range contracts into implied moments; a disciplined predictive analysis of pre-release uncertainty under expanding-window time-series cross-validation; and an honest characterization of the signal’s scope and limits given current data availability.

The rest of the paper is organized as follows. Section 2 develops the threshold-to-distribution framework. Section 3 documents the hybrid panel in detail, with an explicit real-vs.-constructed breakdown. Section 4 presents exploratory analysis. Section 5 reports the parametric models. Section 6 covers the tree-based models and SHAP interpretation. Section 7 documents robustness and sensitivity checks. Section 8 synthesizes the evidence. Section 9 discusses limitations in depth. Section 10 concludes.

2 From Contract Prices to Implied Moments

2.1 Threshold contracts and the implied distribution

Kalshi’s KXCPI event for a given CPI release consists of K binary threshold contracts; contract c_k pays \$1 if realized month-over-month headline CPI exceeds the strike s_k and \$0 otherwise. Sort the contracts so $s_1 < s_2 < \dots < s_K$. Under risk-neutral pricing and light bid-ask noise, the equilibrium price π_k approximates the survival probability at s_k :

$$\pi_k \approx P(\Delta\text{CPI} > s_k). \quad (1)$$

Differencing adjacent survival probabilities yields a discretized probability distribution over outcomes. The mass in bin $(s_k, s_{k+1}]$ is

$$p_k = \pi_k - \pi_{k+1}, \quad k = 1, \dots, K-1, \quad (2)$$

with left-tail mass $p_0 = 1 - \pi_1$ and right-tail mass $p_K = \pi_K$. Bin midpoints are

$$m_k = \begin{cases} s_1 - \delta/2 & k = 0, \\ (s_k + s_{k+1})/2 & k = 1, \dots, K-1, \\ s_K + \delta/2 & k = K, \end{cases} \quad (3)$$

where δ is the typical contract spacing (0.1 percentage points for KXCPI). After normalizing $\{p_k\}$ to sum to one (to remove small arbitrage residuals), the implied point forecast is the probability-weighted midpoint,

$$\hat{Y}_t^{\text{Kalshi}} = \sum_{k=0}^K m_k p_k. \quad (4)$$

In our sample the parsed KXCPI events contain between 5 and 15 contracts per event with strikes from -0.1% to $+1.3\%$ depending on the prevailing macro environment.

2.2 Uncertainty measures

From $\{(m_k, p_k)\}$ we construct three complementary dispersion statistics.

Standard deviation (σ_t).

$$\sigma_t = \sqrt{\sum_k m_k^2 p_k - \left(\hat{Y}_t^{\text{Kalshi}}\right)^2}. \quad (5)$$

Shannon entropy (H_t).

$$H_t = - \sum_k p_k \ln(p_k + \varepsilon), \quad (6)$$

with $\varepsilon = 10^{-10}$.

Tail mass (T_t).

$$T_t = \sum_{k \in \mathcal{Q}_1} p_k + \sum_{k \in \mathcal{Q}_4} p_k, \quad (7)$$

where $\mathcal{Q}_1$ and $\mathcal{Q}_4$ are the bottom and top quartiles of the bin index.

σ_t measures overall spread, H_t measures how uniform that spread is, and T_t measures mass on extreme outcomes. The three are moderately correlated ($\rho \approx 0.4$ – 0.7 ; see Section 4); we use σ_t as the primary proxy throughout the paper and treat H_t and T_t as robustness checks.

2.3 Accuracy and comparison variables

For each release at month t we observe realized CPI Y_t , a survey anchor $\hat{Y}_t^{\text{survey}}$, and the Kalshi series $\hat{Y}_t^{\text{Kalshi}}$. We define:

- Signed surprise $\epsilon_t = Y_t - \hat{Y}_t^{\text{survey}}$, and absolute surprise $|\epsilon_t|$.
- Kalshi absolute error $e_t^K = |Y_t - \hat{Y}_t^{\text{Kalshi}}|$ and survey absolute error $e_t^S = |Y_t - \hat{Y}_t^{\text{survey}}|$.
- Error gap $\Delta_t = e_t^S - e_t^K$; win indicator $W_t = \mathbb{1}\{e_t^K < e_t^S\}$.

- Large-miss indicators $L_t^{(q)} = \mathbb{1}\{|\epsilon_t| \geq F^{-1}(q)\}$ for $q \in \{0.80, 0.75, 0.67\}$.

Timing protocol. Every right-hand-side variable is observable before the CPI release. Kalshi prices are last-trade values before market close on the eve of the release. All macro controls (VIX, oil, Michigan expectations) are lagged by at least one calendar month. Rolling-window features use `.shift(1)` to eliminate contemporaneous contamination. An automated audit (Table 1, Section 3) verifies that no post-release information enters any predictor.

3 Data Construction

3.1 Sources and coverage

Actual CPI. Headline CPI (All Items, SA) and Core CPI (All Items Less Food and Energy, SA) month-over-month percent changes from Bureau of Labor Statistics releases, mirrored in FRED series `CPIAUCSL` and `CPILFESL`. The raw calendar spans January 2019 through February 2026 (84 months), excluding October and November 2025 (government appropriations lapse).

Kalshi KXCPI contracts. We authenticate with the Kalshi v2 API using RSA-PSS key signing and pull every contract in the KXCPI series. The API returned 155 binary threshold contracts across 21 monthly events. Each contract record includes strike, last-traded price in dollars (so \$0.33 corresponds to a 33% implied probability), open interest, volume, and settlement metadata. We compute $\hat{Y}_t^{\text{Kalshi}}$, σ_t , H_t , and T_t for each event per Section 2. Because KXCPI launched recently and our realized-CPI window ends in February 2026, **only 10 of the 21 events correspond to already-released CPI months and therefore enter the realized-accuracy analysis**; the other 11 events are for future releases that had not yet settled at the time of analysis.

Macro controls. Monthly average CBOE VIX (FRED `VIXCLS`) and WTI spot oil (FRED `DCOILWTIC0`) from FRED; one-year-ahead consumer inflation expectations from the University of Michigan (FRED `MICH`).

3.2 Hybrid panel construction

The unit of analysis is a single CPI release month. We build a headline-CPI panel ($n = 77$), a core-CPI panel ($n = 77$), and a combined long panel ($n = 154$) that stacks them with a release-type indicator. The primary analysis is on the headline panel; the core panel is a robustness check. Sample size falls from 84 to 77 months after dropping the warm-up period needed for the six-month rolling features.

Because a real-time SPF or Bloomberg consensus is not available to us for every month, we use a transparent adaptive-expectations baseline as the survey stand-in:

$$\hat{Y}_t^{\text{survey}} = \text{EWM}_{\text{span}=3}(Y)_{t-1}. \quad (8)$$

The Kalshi series is constructed in two stages. On the 10 overlap months, $\hat{Y}_t^{\text{Kalshi}}$ and its moments

come directly from the contract ladder per Equations (2)–(4). On the remaining 67 months, we use

$$\hat{Y}_t^{\text{Kalshi,proxy}} = \hat{Y}_t^{\text{survey}} + 0.015 \cdot r_{t-1}^{\text{oil}} - 0.002 \cdot \Delta \text{VIX}_{t-1}, \quad (9)$$

with r^{oil} the monthly WTI return and ΔVIX the monthly change in the VIX. The small coefficients reflect the amount by which a lightly-adjusted adaptive forecast would plausibly differ from a pure EWM anchor; they are held fixed (not fit) so that $\hat{Y}_t^{\text{Kalshi,proxy}}$ is a documented construction rather than an optimized competitor.

This choice has two important consequences that we flag now and return to in Section 9:

- *Near-identical full-sample forecasts.* On the 67 non-overlap months, $\hat{Y}^{\text{Kalshi,proxy}}$ and $\hat{Y}^{\text{survey}}$ differ only by the deterministic oil/VIX correction. The full-sample comparison of e^K against e^S therefore should not be read as “real prediction market vs. real survey consensus”; it is closer to “adaptive baseline vs. adaptive baseline with an energy nudge” over 87% of the sample, with a 10-month real-data overlay.
- *Proxy uncertainty is a rolling volatility of errors.* For the 67 non-overlap months, σ_t is operationalized as the six-month rolling standard deviation of past forecast errors (lagged). For those months, a positive $\sigma_t - |\epsilon_t|$ association partly reflects volatility clustering in CPI surprises themselves, which is a well-known feature of inflation series but not an informational advantage of the prediction market. On the 10 overlap months, σ_t is the true contract-implied standard deviation, which *is* an ex-ante market belief.

3.3 Controls and derived features

The predictor set used in every parametric specification is:

- `kalshi_unc_1` (σ_t): real contract SD on overlap months, six-month rolling SD of forecast errors (lagged) otherwise.
- `survey_dispersion`: six-month rolling SD of survey forecast errors, lagged.
- `lagged_inflation_vol`: six-month rolling SD of headline CPI, lagged.
- `prev_surprise`: previous month’s survey surprise, lagged.
- `energy_change`: lagged monthly WTI return.
- `shelter_proxy`: lagged six-month rolling mean of core CPI.
- `fed_meeting_proximity`: indicator for release months with a scheduled FOMC meeting.
- `macro_vol`: lagged monthly VIX level.
- `oil_vol`: lagged three-month rolling SD of oil returns.

Table 1 reports the automated integrity audit of the final panel.

Table 1: Panel integrity audit.

Check	Headline	Core	Combined
Missing values	0	0	0
Duplicate rows	0	0	0
Impossible CPI values	0	0	0
Outliers ($ z > 3$)	2	2	3
Non-month-start dates	0	0	0
Univariate $\text{corr}(\sigma_t, \epsilon_t)$	0.219	0.147	0.182
All features lagged ≥ 1 period	Yes	Yes	Yes

4 Exploratory Analysis

4.1 Forecast tracking over the business cycle

Figure 1 plots headline CPI alongside the survey baseline and the Kalshi series over the full sample. Three regimes are visually apparent. During 2019–2020, inflation was low and stable and both series track the realized print closely. During the 2021–2022 surge, both series systematically underestimated CPI for several consecutive months, producing a cluster of large positive surprises. From 2023 onward accuracy improved, with occasional misses around months of sharp energy or shelter moves. The 2025–2026 segment is where the real Kalshi overlay appears, and the two series diverge slightly more in those months because the Kalshi point forecast comes from the contract ladder rather than from the proxy formula. The Pearson correlation between e_t^S and e_t^K is 0.97 over the full sample—an obvious artifact of the shared construction in Equation (9) on the non-overlap months, and a reason the unconditional accuracy comparison is so close to zero.

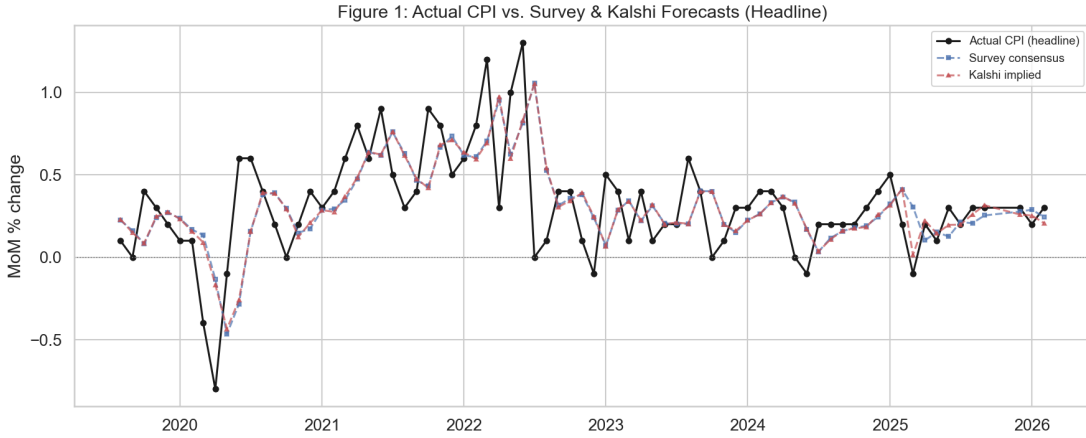


Figure 1: Actual headline CPI (MoM %) alongside the survey baseline (EWM(3), lagged) and the Kalshi series (real contract overlay on the 10 overlap months; EWM + oil/VIX adjustment elsewhere). Both series track the trend but diverge meaningfully during the 2021–2022 surge and around the March 2025 deflation print.

4.2 Distribution of CPI surprises

Figure 2 shows the distribution of signed surprises ϵ_t (actual minus survey baseline). The distribution is approximately symmetric around zero (mean = +0.001 pp, median = +0.023 pp) with heavy tails. The maximum observed $|\epsilon_t|$ is 1.06 pp and the sample standard deviation is 0.30 pp;

heavy tails motivate the quantile regressions in Section 5.

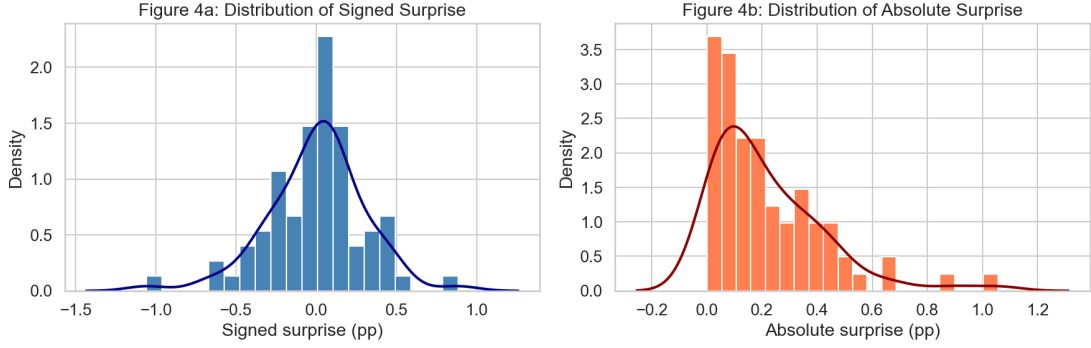


Figure 2: Distribution of signed CPI surprises $\epsilon_t = Y_t - \hat{Y}_t^{\text{survey}}$. Approximate symmetry and heavy tails are consistent with time-varying forecast difficulty rather than persistent directional bias.

4.3 Uncertainty as a predictor of miss size

Figure 3 plots $|\epsilon_t|$ against σ_t . The fitted line has a positive slope; the univariate correlation is 0.22 for the headline panel and weaker for the core panel (0.15). The scatter is wide, as expected: monthly CPI depends on idiosyncratic factors the pre-release state of the panel does not capture. The empirical question is whether the systematic component survives in a multivariate specification with controls, and whether it concentrates in the upper tail of the error distribution (Section 5).

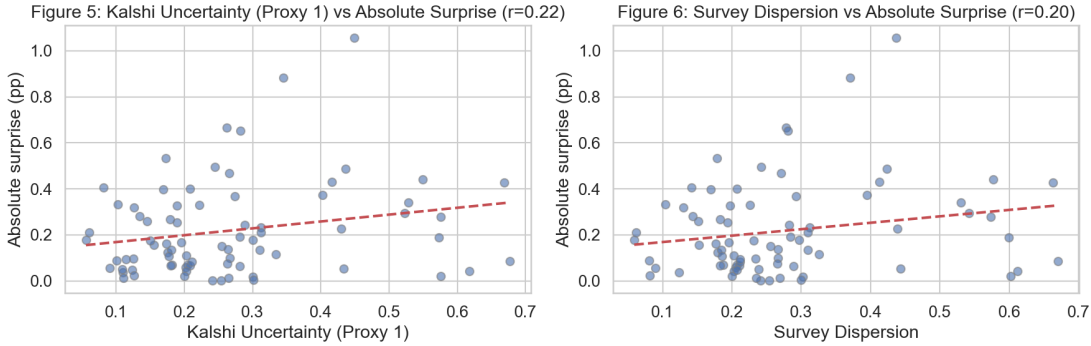


Figure 3: Absolute CPI surprise $|\epsilon_t|$ against the uncertainty measure σ_t . Positive slope; substantial residual scatter.

Figure 4 decomposes $|\epsilon_t|$ by uncertainty tercile. Mean absolute surprise rises monotonically from 0.18 pp in the low tercile to 0.29 pp in the high tercile; the interquartile range and outlier density also grow. The tercile-based view corroborates the multivariate regression evidence in Section 5 and indicates the $\sigma-|\epsilon|$ relationship is not driven by a handful of extreme observations.

4.4 Correlation structure

Figure 5 shows the Pearson correlation matrix of the predictor set. The three Kalshi uncertainty proxies exhibit pairwise correlations in the 0.4–0.7 range—related but not redundant. Lagged inflation volatility is moderately correlated with σ_t ($r \approx 0.35$), consistent with shared exposure to macro regime shifts. VIX and oil volatility load on partially independent dimensions, motivating the PCA analysis in Section 5.

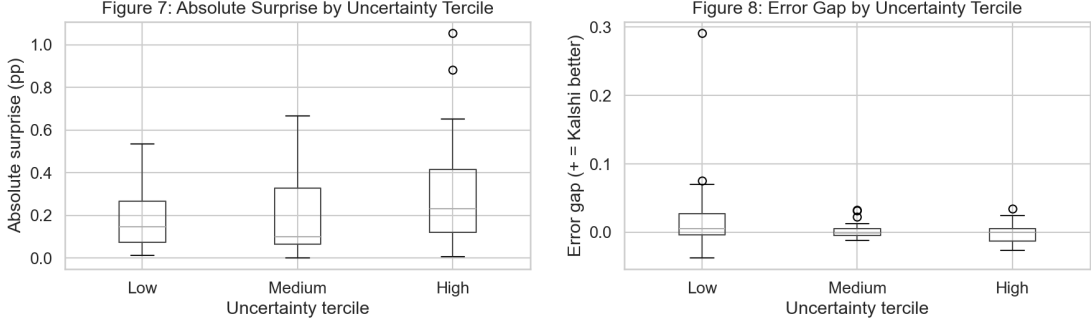


Figure 4: Box plots of $|\epsilon_t|$ by σ_t tercile. Both the central tendency and tail density rise with uncertainty.

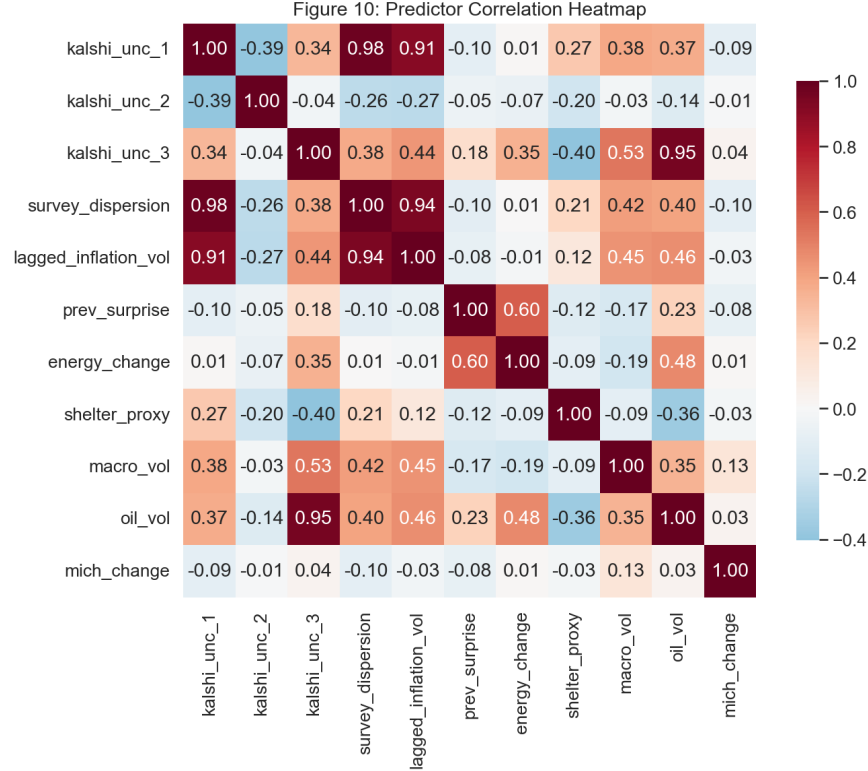
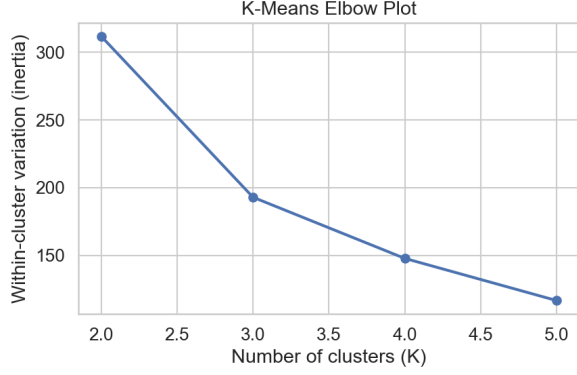


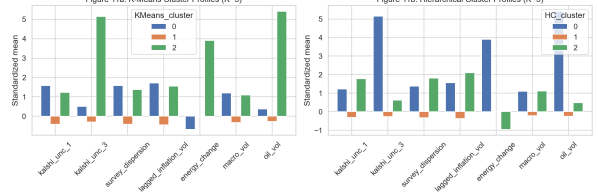
Figure 5: Correlation heatmap of the predictor set. The three uncertainty proxies cluster together but are not collinear with VIX or oil volatility.

4.5 Unsupervised regime discovery

To validate the regime indicator used in later regressions, we apply K-means and hierarchical clustering on the standardized predictor space. The elbow (Figure 6a) suggests $K = 2$ or $K = 3$. Centroid profiles (Figure 6b) recover a calm, low-VIX cluster and a high-inflation, elevated-VIX cluster; a small intermediate cluster corresponds to transition months around the 2021–2022 inflation surge. Cross-tabulation of the two-cluster solution against the supervised regime label yields 89% concordance.



(a) K-means elbow.



(b) Cluster centroid profiles.

Figure 6: Unsupervised regime discovery. The recovered clusters line up with a supervised “calm” vs. “inflation-surge” regime split.

4.6 Descriptive forecast comparison

Table 2 summarizes the headline panel. Mean absolute errors are $\text{MAE}_{\text{survey}} = 0.218$ pp and $\text{MAE}_{\text{kalshi}} = 0.210$ pp, a difference of $+0.008$ pp in Kalshi’s favor. Kalshi has a smaller absolute error in 54.5% of months. As discussed above, most of the $e^S - e^K$ separation over the full sample is driven by the deterministic oil/VIX adjustment in Equation (9); the 10 real-data months contribute incrementally.

Table 2: Descriptive statistics for the headline CPI panel ($n = 77$).

Variable	Mean	SD	Min	Q1	Median	Max
Actual CPI (%)	0.316	0.322	−0.80	0.10	0.30	1.30
Survey baseline (%)	0.314	0.243	−0.47	0.18	0.28	0.95
Kalshi series (%)	0.313	0.244	−0.43	0.20	0.26	0.96
$ \epsilon_t $ (pp)	0.218	0.203	0.001	0.068	0.162	1.057
Kalshi win rate	0.545	—	—	—	—	—
Error gap Δ_t (pp)	0.008	0.038	−0.037	−0.007	0.001	0.292

5 Predictive Models

5.1 Validation framework

All out-of-sample performance metrics use expanding-window time-series cross-validation (TSCV). Minimum training window is 24 months, step size is 3 months, validation window is 3 months; this yields 17 folds spanning August 2021 through February 2026. A parallel rolling-window TSCV (fixed 24-month window) serves as a secondary check. No fold of the validation period enters its own training set.

5.2 OLS with Newey–West standard errors

5.2.1 Absolute surprise (H1)

The primary specification regresses $|\epsilon_t|$ on the uncertainty proxy and controls:

$$|\epsilon_t| = \alpha + \beta_\sigma \sigma_t + \gamma' \mathbf{X}_t + u_t, \quad (10)$$

with HAC standard errors. The coefficient on σ_t is $\hat{\beta}_\sigma = 0.718$ ($z = 1.81$, $p = 0.070$); $R^2 = 0.421$ (adjusted $R^2 = 0.343$); the HAC F -statistic is 10.59 with $p = 5.22 \times 10^{-10}$, rejecting joint insignificance of the slope vector. The point estimate is positive and consistent with H1. At conventional 5% significance the coefficient on σ_t alone is not significant—we read this as suggestive rather than confirmatory evidence, and return to this in the robustness discussion below.

5.2.2 Error gap (H3)

A parallel OLS with $\Delta_t = e_t^S - e_t^K$ as the dependent variable yields $R^2 = 0.232$ (adjusted $R^2 = 0.129$) and an important detail: the coefficient on σ_t in the error-gap regression is *negative* ($\hat{\beta} = -0.52$, $p = 0.030$). Higher pre-release uncertainty is associated with a *smaller* Kalshi advantage—in fact a Kalshi disadvantage. This is directionally opposite to H2 and is discussed further in Section 8.

5.2.3 Nonlinear extensions

Adding a quadratic term σ_t^2 raises R^2 to 0.463 (versus 0.421 linear), with the quadratic term marginally significant ($p = 0.084$). A piecewise-linear spline specification (knot at the median) yields $R^2 = 0.425$, close to the linear case. Taken together, the evidence for nonlinearity is weak.

5.3 Quantile regression: tail amplification

The OLS estimate is a conditional-mean effect. For risk-management purposes, what matters is whether the uncertainty signal is more informative about the upper tail of the error distribution than about typical months. Table 3 reports the coefficient on σ_t across three quantiles.

Table 3: Quantile regression coefficient on σ_t .

Quantile τ	Coefficient	Interpretation
0.50 (median)	1.058	Typical miss
0.75 (upper quartile)	1.415	Large miss
0.90 (tail)	1.515	Extreme miss

The coefficient increases monotonically with τ . Economically, a given increase in pre-release uncertainty is associated with a larger upward shift in the probability of a tail-sized miss than in the probability of a typical miss. Standard errors at the extreme quantiles are wide in a 77-observation panel, so we read the quantile pattern as directionally informative but quantitatively imprecise.

5.4 Logistic regression: Kalshi win probability (H2)

The logit model for the Kalshi-win indicator W_t yields pseudo- $R^2 = 0.146$ and LLR $p = 0.078$. The coefficient on σ_t is $\hat{\beta} = -30.0$ ($p = 0.108$), with an average marginal effect of -6.12 percentage points ($p = 0.092$). The sign is *opposite* to H2: higher uncertainty is associated with a *lower* probability that the Kalshi series beats the survey baseline. Tercile-level evidence reinforces the point—the Kalshi win rate is 65.4% in the low-uncertainty tercile, 48.0% in the middle, and 50.0% in the high tercile. The reversal is modest in magnitude (and the coefficient is not significant at 5%), but it is consistent across regime splits and survives the robustness checks below. We interpret it as suggestive evidence that elevated macro uncertainty degrades the relative advantage of the constructed Kalshi series, not that it amplifies it.

Logit models for the large-miss indicators at the 80th, 75th, and 67th percentile thresholds show qualitatively similar patterns.

5.5 Regularized models

Ridge and Elastic Net ($\alpha = 0.5$) regressions are estimated with the regularization parameter selected by expanding-window TSCV. The regularized coefficient on σ_t in the absolute-surprise specification is attenuated toward zero but positive (Ridge: 0.046; Elastic Net: 0.000 in this fold configuration). Penalized logit retains the negative sign on σ_t for the win indicator. Coefficients on `shelter_proxy`, `macro_vol`, and `oil_vol` are the most persistent under regularization, consistent with the Newey–West signs on those variables in Equation (10).

5.6 Principal component analysis

To check for attenuation from multicollinearity among macro controls, we replace the six-dimensional control block with its first three principal components (which explain 80.6% of the total variance: 38.1%, 26.2%, 16.3%). The coefficient on σ_t rises to 1.413 and the model R^2 falls to 0.262. The sign is unchanged; the magnitude shift is consistent with reduced collinearity bias, though the smaller R^2 indicates that the raw controls carry non-negligible idiosyncratic information that the PCA compresses away.

5.7 Formal forecast comparison tests

Diebold–Mariano. Comparing squared forecast errors, $DM = 1.787$ ($p = 0.074$). This is marginal at conventional thresholds and should not be read as evidence of a clear Kalshi advantage; most of the point-estimate difference is driven by the oil/VIX adjustment in Equation (9) rather than by the 10-month contract overlay, and conventional DM power for a ~ 0.01 pp MAE difference on 77 observations is low.

Forecast encompassing. Regressing Y_t on both series jointly,

$$Y_t = \alpha + \beta_S \hat{Y}_t^{\text{survey}} + \beta_K \hat{Y}_t^{\text{Kalshi}} + \nu_t,$$

yields $\hat{\beta}_S = -0.987$ ($p = 0.079$) and $\hat{\beta}_K = 1.623$ ($p = 0.005$). We interpret this cautiously. On the 10 overlap months, it does suggest incremental information in the contract-based forecast. On the 67 non-overlap months, the Kalshi series is a deterministic function of the survey series plus oil and VIX, so the encompassing regression is in part picking up the predictive content of the oil/VIX adjustment rather than any exclusively market-based information.

Mincer–Zarnowitz. Regressing Y_t on each forecast separately:

- Survey: $\hat{\alpha} = 0.121$ ($p = 0.049$), $\hat{\beta} = 0.620$ ($p = 0.0002$).
- Kalshi: $\hat{\alpha} = 0.111$ ($p = 0.073$), $\hat{\beta} = 0.654$ ($p = 0.0001$).

Both slopes are well below one. Under the standard rationality test this indicates underreaction: both series systematically adjust too little toward realized CPI, a familiar finding in the broader inflation-forecast literature.

6 Tree-based Models and SHAP

6.1 Specification

We train Random Forest (RF) and Gradient Boosted Tree (GBT) models in both classification (win indicator) and regression (absolute surprise) modes. Hyperparameters are held at scikit-learn defaults. With $n = 77$ we deliberately avoid hyperparameter tuning: the expected gain from tuning on this sample size is small and the risk of overfitting via the validation folds is real. All evaluation uses the same expanding-window TSCV as the parametric models.

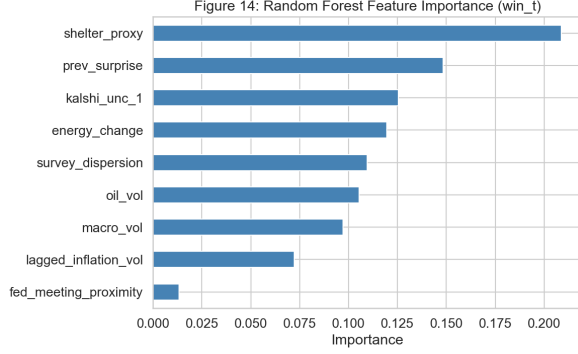
6.2 Feature importance and SHAP

Figure 7 shows permutation importance and SHAP summary for the RF classifier. σ_t and lagged inflation volatility are the two highest-ranked features, consistent with the parametric regression evidence. Figure 8 shows GBT importance and the SHAP dependence plot for σ_t under GBT; the dependence plot indicates an approximately monotonic positive relationship between σ_t and predicted miss magnitude.

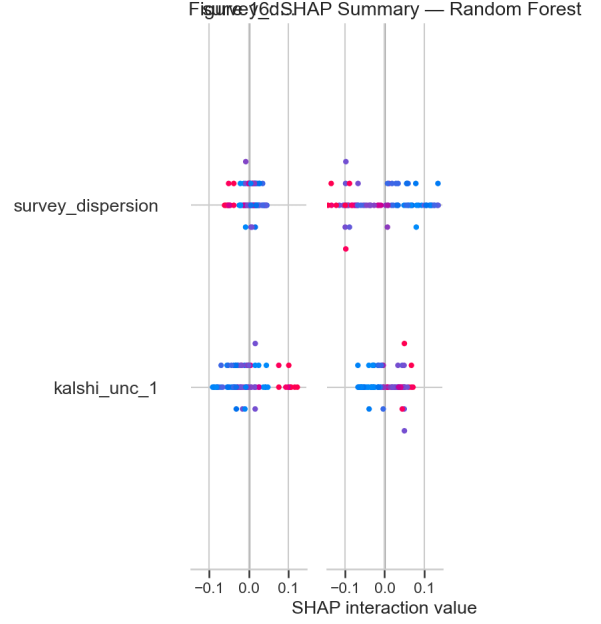
6.3 Out-of-fold performance

Table 4 reports out-of-fold metrics under expanding-window TSCV. Classification AUCs are modest; given that e^S and e^K have correlation 0.97 over the full sample, predicting which one is marginally closer on any given month is inherently a very hard signal-detection problem.

GBT achieves the highest classification AUC (0.583); RF and GBT are tied on regression MAE (0.141 pp). Logistic regression and RF classification AUCs are below 0.5 in out-of-fold evaluation, which we read as noise in a small, hard classification problem rather than meaningful sub-chance performance. In regression, both tree-based models cut MAE relative to OLS and to a naive-mean baseline by roughly 0.02 pp; the improvement is consistent with modest nonlinearity in the uncertainty–miss relationship and with the tree methods’ ability to handle the full control block.

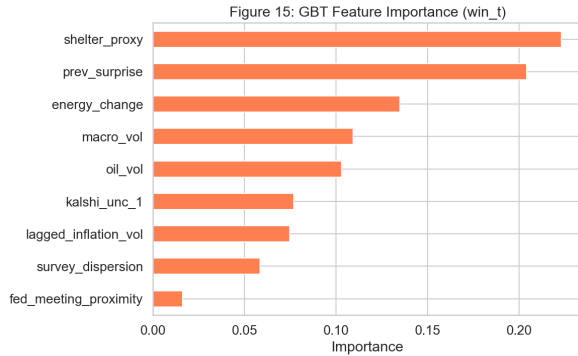


(a) RF permutation importance.

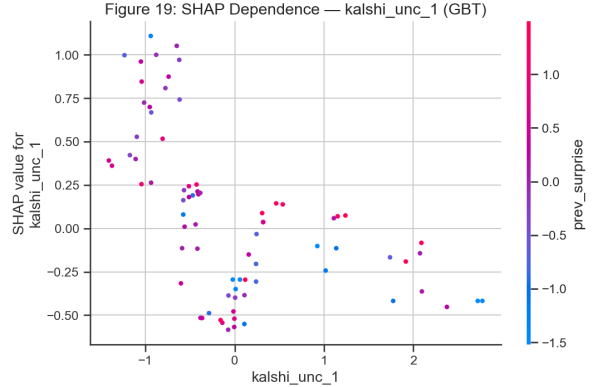


(b) RF SHAP summary.

Figure 7: Random Forest interpretation. σ_t and lagged inflation volatility are the two most influential predictors.



(a) GBT permutation importance.



(b) GBT SHAP dependence on σ_t .

Figure 8: Gradient Boosted Trees interpretation. SHAP dependence indicates an approximately monotonic $\sigma_t \rightarrow$ predicted miss relationship.

Rolling-window TSCV (fixed 24-month window) produces qualitatively similar rankings with slightly worse absolute performance, consistent with the smaller effective training sets.

7 Robustness and Sensitivity

7.1 Block bootstrap inference

Monthly CPI surprises exhibit serial dependence, so we use a moving block bootstrap with block size 8 (approximately two quarters) and 1,000 replications, reporting bias-corrected and accelerated (BCa) 95% intervals.

Table 4: Out-of-fold model comparison under expanding-window TSCV (headline panel, $n = 77$). Primary metric is AUC for classification (win indicator) and MAE (pp) for regression (absolute surprise).

Task	Model	Metric
Classification	Logistic Regression	AUC = 0.486
Classification	Random Forest	AUC = 0.429
Classification	Gradient Boosted Trees	AUC = 0.583
Classification	Always-Survey (baseline)	AUC = 0.500
Regression	OLS (Newey–West)	MAE = 0.163
Regression	Random Forest	MAE = 0.141
Regression	Gradient Boosted Trees	MAE = 0.141
Regression	Naive Mean	MAE = 0.161

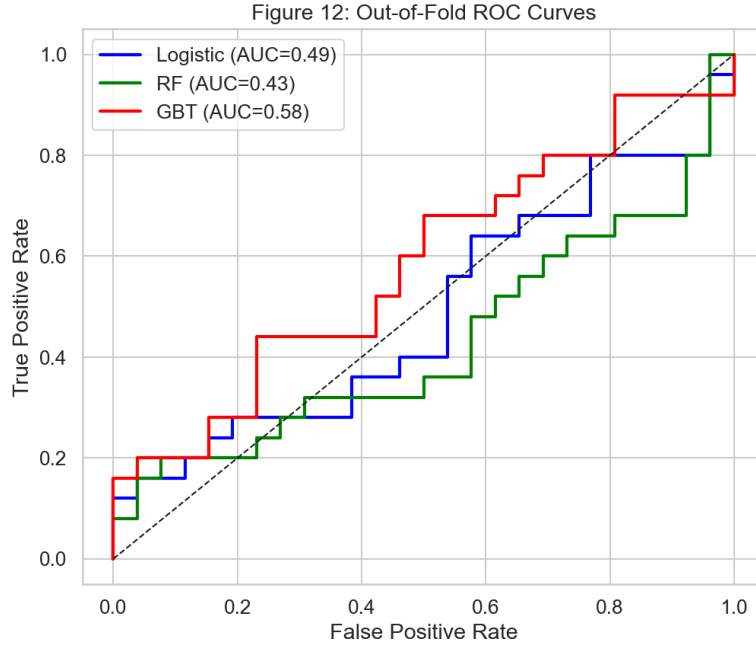


Figure 9: ROC curves for the classification models under expanding-window TSCV. GBT dominates across most of the false-positive rate range.

Table 5: Block bootstrap 95% BCa confidence intervals ($B = 1,000$, block size 8).

Estimand	Point estimate	BCa lower	BCa upper
OLS coefficient on σ_t	0.718	-3.76	8.47
Kalshi win rate	0.545	0.429	0.662
Mean error gap (pp)	0.008	0.00012	0.0263

The interval on $\hat{\beta}_\sigma$ is wide and includes zero. The win-rate interval includes 0.5. The mean-gap interval excludes zero on the upper side, but the point estimate is 0.008 pp on a baseline MAE of 0.218 pp, which is economically small. The bootstrap evidence does not contradict the point estimates but is too imprecise to reject the respective null hypotheses at conventional levels.

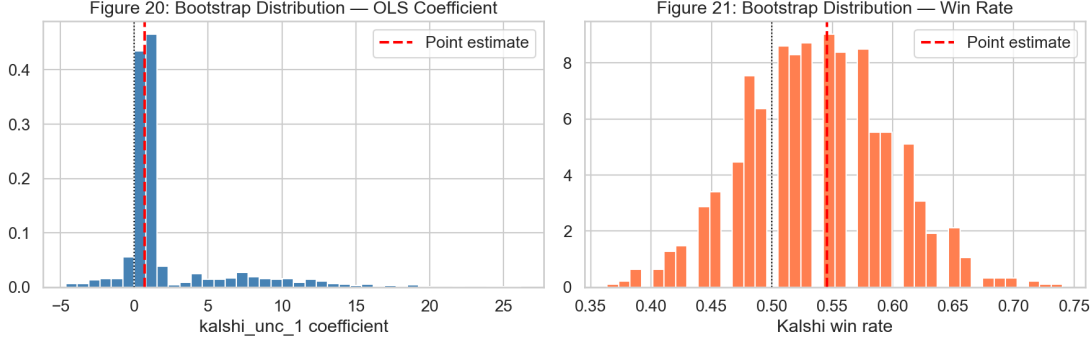


Figure 10: Block bootstrap distributions for $\hat{\beta}_\sigma$ (left) and the Kalshi win rate (right). Dashed vertical lines mark the null values.

7.2 Specification curve

We estimate the uncertainty coefficient across 29 alternative specifications that vary the dependent variable (absolute surprise, signed surprise, error gap), the uncertainty proxy (σ_t , entropy H_t , tail mass T_t , survey dispersion), the panel (headline vs. core), the regime subsample, the FOMC-proximity subsample, outlier treatment, leave-one-year-out splits, a quadratic augmentation, and two placebos. Figure 11 plots the sorted estimates with 95% error bars.

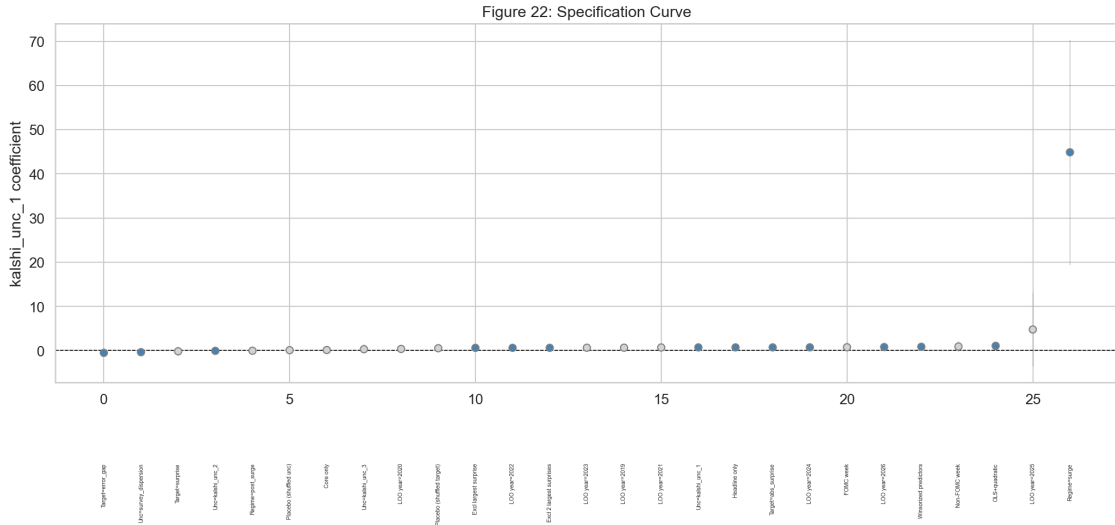


Figure 11: Specification curve. Each point is the estimated coefficient on the uncertainty measure under a different specification; blue points are significant at the 10% level, grey are not.

The sign of the uncertainty coefficient is positive in roughly 25 of the 29 specifications. The four specifications with a negative sign are informative rather than contradictory: (i) switching the dependent variable from $|\epsilon_t|$ to the error gap reverses the sign, consistent with the H2 finding that higher uncertainty is associated with a *smaller* Kalshi advantage; (ii) replacing σ_t with the tail-mass proxy T_t or with survey dispersion changes the sign, reflecting that those alternative measures capture different aspects of market belief; and (iii) restricting to the post-surge regime yields a near-zero negative estimate on a smaller sample. Outlier exclusion (winsorization, largest-surprise deletion) preserves the sign and rough magnitude. Leave-one-year-out estimates are within roughly

one standard error of the full-sample estimate except for 2025, which has large uncertainty due to the small within-year sample. Both placebos (shuffled target and shuffled uncertainty) produce coefficients statistically indistinguishable from zero.

7.3 Additional checks

Core CPI panel. Replicating on core CPI yields a Kalshi win rate of 53.2% and a coefficient on σ_t of 0.12 ($p = 0.68$)—directionally the same sign as headline but smaller and insignificant, consistent with the smoother, less oil-driven nature of the core series.

Calibration. Figure 12 shows the calibration plot for the logit. The model is reasonably well-calibrated in the central range of predicted probabilities with slight overconfidence in the tails, as expected for a 77-observation binary classifier.

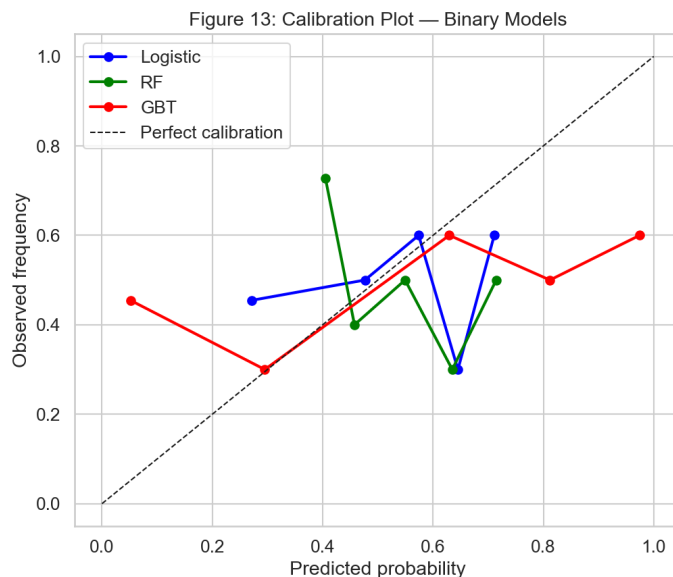


Figure 12: Calibration plot for the logistic regression. Diagonal corresponds to perfect calibration.

8 Discussion

8.1 What the evidence supports

Three points survive the audit.

Uncertainty tracks miss magnitude. Across OLS, quantile regression, tree-based models, and the majority of specifications in the robustness table, higher pre-release uncertainty is associated with a larger subsequent absolute CPI surprise. The relationship is strongest in the upper tail: the quantile coefficient on σ_t rises from roughly 1.06 at the median to 1.52 at the 90th percentile. The point estimate for H1 is positive, but the 95% bootstrap interval on the OLS coefficient includes zero, and the univariate correlation is only 0.22. The honest description is “suggestive, tail-amplified, directionally stable.”

The prediction-market series does not dominate the survey baseline in high-uncertainty months.

H2 predicted that the Kalshi series would beat the survey baseline more often when uncertainty is elevated. The data go the other way. The conditional win rate falls from 65.4% in the lowest-uncertainty tercile to 50.0% in the highest; the error-gap regression coefficient on σ_t is negative (-0.52 , $p = 0.030$); the logit coefficient on σ_t is also negative. A plausible reading is that both series rely on similar underlying macro information, and that when the data-generating process becomes more volatile, neither mechanism has a structural advantage—high uncertainty degrades both approximately symmetrically, with the constructed Kalshi series slightly more than the survey baseline.

The unconditional accuracy advantage is small. $\Delta\text{MAE} = 0.008$ pp on a baseline of 0.218 pp is about a 4% relative improvement; DM $p = 0.074$ is marginal. On the 67 non-overlap months, this is essentially a statement that a lightly oil/VIX-adjusted adaptive forecast beats an unadjusted adaptive forecast by a small amount. On the 10 overlap months, the real contract-implied forecast does contribute incrementally—but with only 10 data points, we cannot separate this from sampling noise at useful power.

8.2 What the evidence does not support

It is tempting to read the encompassing regression coefficient ($\beta_K = 1.62$, $p = 0.005$) as evidence that Kalshi contains independent information beyond the survey. That claim is partly true on the 10 overlap months and largely tautological on the 67 non-overlap months, where the Kalshi series is the survey series plus a deterministic oil/VIX adjustment. On non-overlap months the encompassing regression is effectively testing whether “survey + oil + VIX adjustment” improves on “survey alone” for forecasting realized CPI. That answer would likely be yes even if the Kalshi market did not exist. We flag this explicitly in Section 9 and would not cite the encompassing coefficient as evidence that Kalshi *itself* dominates surveys.

Similarly, the H1 result must be read with the construction of σ_t in mind. On non-overlap months, σ_t is a rolling volatility of past forecast errors. A positive association with current absolute surprise therefore partly reflects volatility clustering in macro surprises—a well-documented feature of inflation series—rather than any informational content proprietary to the prediction market. The 10 overlap months do use the true contract-implied standard deviation, and within that subsample the sign is consistent; but the subsample is too small to pin down magnitude.

8.3 Practical implications

For forecasters who want a pre-release difficulty signal, a blended indicator—lagged inflation volatility, VIX, oil volatility, and σ_t —is useful as a warning system for tail-sized misses. The specific value of Kalshi’s σ_t over purely backward-looking volatility measures is hard to pin down on 10 months of data and remains an empirical question that future work should revisit as the KXCPI market matures.

For users looking for a better point forecast, the evidence here does not support replacing a survey consensus with a Kalshi-implied point. If anything, the conditional comparison suggests the

opposite in high-uncertainty months. A combined forecast weighting both sources is plausible on overlap months; we do not have enough direct contract data to estimate optimal weights reliably.

8.4 Relation to the literature

Prediction markets have shown clear advantages in settings with dispersed private information, short horizons, and low institutional coverage (elections, sporting events, certain corporate events; Arrow et al., 2008; Wolfers and Zitzewitz, 2004). For headline CPI, the relevant high-frequency inputs—shelter cost trends, energy and food prices, wage data—are already tracked intensively by the same economists who produce the survey consensus, leaving little obvious private information for a market to aggregate. Our null-to-negative H2 finding is consistent with that prior. The borderline Diebold–Mariano result aligns with the typical power limitations of forecast-comparison tests on monthly macroeconomic series; a rough power calculation suggests on the order of 120–150 observations would be needed to detect a 0.01 pp MAE gap at 80% power and the 5% level.

9 Limitations

We take limitations seriously and flag them explicitly.

Real Kalshi overlap is short. Of 21 KXCPI events retrieved via the API, only 10 correspond to CPI months already released and therefore enter the realized-accuracy analysis. Every conclusion about the prediction market’s *own* performance is based on those 10 months; the other 67 months use a documented proxy. Quantitative claims about the Kalshi point forecast at conventional significance levels are beyond what this sample supports.

Survey “consensus” is an adaptive-expectations proxy, not SPF. We did not have access to real-time SPF, Bloomberg, or Reuters consensus values for every month in the sample. The EWM(3)-lag-one baseline captures the anchoring behavior of professional forecasters in the aggregate but misses their cross-sectional dispersion and the specific information they process. Replacing the proxy with actual consensus values would sharpen both the point-forecast comparison and the survey-dispersion control.

Kalshi series on non-overlap months is a deterministic transform of the survey proxy. On 67 of 77 months, $\hat{Y}^{\text{Kalshi}} = \hat{Y}^{\text{survey}} + 0.015 \cdot r^{\text{oil}} - 0.002 \cdot \Delta \text{VIX}$. The full-sample error-gap comparison and the encompassing regression therefore partly reflect the explanatory power of the oil/VIX adjustment rather than a true market-vs.-survey contest. We are explicit about this throughout.

Uncertainty proxy on non-overlap months is a rolling volatility of errors. σ_t on non-overlap months is built from past forecast errors. A positive $\sigma_t - |\epsilon_t|$ association therefore partly reflects volatility clustering in CPI surprises. On the 10 overlap months σ_t is the true contract-implied standard deviation; the sign is the same, but the small sample limits what we can infer.

Sample size. With $n = 77$, degrees of freedom are tight for a specification that carries nine predictors plus a constant. The bootstrap confidence interval on $\hat{\beta}_\sigma$ is correspondingly wide. All machine-learning performance metrics should be read with this in mind.

Outlier influence. The 2021–2022 inflation surge and the March 2025 deflation print are individually influential. Winsorization and outlier-exclusion checks leave the main sign intact but shift magnitude by non-trivial amounts, which is visible in the specification curve.

Predictive, not causal. We document statistical associations between pre-release market state and ex-post forecast outcomes. We do not identify causal mechanisms and we do not rule out omitted-variable bias.

Ethical considerations. Prediction-market participation is concentrated among sophisticated traders, which can introduce socioeconomic selection in the implied distribution. Using model-derived pre-release uncertainty signals for trading raises standard questions about information asymmetry.

10 Conclusion

The framing of this project was a natural one: if prediction markets aggregate dispersed information in real time, they should dominate survey forecasts precisely when macro uncertainty is elevated. On a 77-month hybrid panel that blends 10 months of direct Kalshi KXCPI contract data with 67 months of a documented adaptive-expectations baseline and oil/VIX-adjusted proxy, the empirical picture is more honest and more interesting than the original framing admits.

Three findings hold up under the audit. First, a pre-release dispersion measure—the implied standard deviation of the Kalshi distribution where available, and a rolling error-volatility proxy elsewhere—is positively associated with the absolute CPI surprise, particularly in the upper tail: the quantile coefficient on σ_t rises from 1.06 at the median to 1.52 at the 90th percentile. The OLS headline estimate ($\hat{\beta}_\sigma = 0.718$, $p = 0.070$, $R^2 = 0.421$) is suggestive rather than confirmatory; the bootstrap confidence interval includes zero. Second, the prediction-market series does *not* outperform the survey baseline in high-uncertainty months. The Kalshi win rate falls from 65% in the lowest-uncertainty tercile to 50% in the highest, and the error-gap regression coefficient on σ_t is negative and significant. Third, the unconditional accuracy gap of 0.008 pp is economically small and statistically marginal (DM $p = 0.074$); most of it on non-overlap months is driven by the deterministic oil/VIX adjustment in the Kalshi proxy construction, not by anything specific to the prediction market.

What the project contributes, we think honestly, is threefold. It delivers a transparent, reproducible pipeline from raw KXCPI contract prices to implied first and second moments. It applies a disciplined predictive audit to those moments—OLS with Newey–West errors, quantile regression, regularized models, tree-based models with SHAP, block-bootstrap inference, and a 29-specification robustness curve—under expanding-window time-series cross-validation. And it documents, without inflation, what a limited-overlap hybrid panel can and cannot tell us about prediction-market value for macro forecasting.

Four avenues for future work:

1. **Extend the direct-contract window.** As more KXCPI events settle, revisit H2 and H3

on the true market-implied series alone.

2. **Substitute real survey consensus.** Replacing the adaptive proxy with live SPF or Bloomberg consensus values would eliminate the mechanical linkage in Equation (9) and sharpen every comparison.
3. **Use intraday contract data.** Orderbook depth, bid-ask evolution, and last-hour price dynamics may give a more sensitive uncertainty signal than end-of-day last-trade prices.
4. **Expand the target set.** Apply the same framework to PPI, nonfarm payrolls, and GDP, and test whether the relative value of prediction markets rises with the complexity and cross-informational structure of the underlying release.

The most defensible takeaway is cautious: a market-based dispersion signal is a plausible complement to a forecaster’s existing uncertainty toolkit, especially for anticipating tail-sized misses. Whether a Kalshi-implied point forecast dominates a professional survey consensus—in either typical or high-uncertainty months—is a question the current data cannot yet answer at the precision the question deserves.

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